

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

QIAN QI

ABSTRACT. Invariant-integral and pressure null spaces are defined as closed linear spans. Platform-specific Cesaro Lyapunov bounds give invariant Radon separators. Source response is transported on one common transfer-operator Banach bundle by Kato parallel transport, avoiding nonexistent infinite-path Radon–Nikodym densities; the resolved Hilbert projection is constructed separately and the covariant memory and optional-projection diagrams follow on their proper domains.

Platform identifier: TYPED-COPRODUCT-v3. **Controlling revision:** ROUND10-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** C2_COMMON_TRANSFER_BUNDLE_COTANGENTS.tex.

CONTENTS

1. Round-ten reconstruction: closed nullspaces, common transfer bundles, and typed contractions	1
1.1. Two genuine closed linear nullspaces	2
1.2. A common transfer Banach bundle	2
1.3. Resolved memory on the transported subspace	3
1.4. Optional projections and nonlinear contractions	3
Scope and verification status	4

1. ROUND-TEN RECONSTRUCTION: CLOSED NULLSPACES, COMMON TRANSFER BUNDLES, AND TYPED CONTRACTIONS

Every platform retains its label. Cross-platform statements require an explicit scaling map. Within one platform, path potentials live in a weighted dynamical algebra $\mathcal{A}_W$ invariant under time shifts.

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 60F10, 60H10; Secondary 37A50, 49L25.

Key words and phrases. path-space large deviations, cotangent fields, entropic rigidity, tangent laws, Girsanov, BSDE.

1.1. Two genuine closed linear nullspaces. Define

$$\begin{aligned}\mathcal{N}_{\text{int}} &= \overline{\text{span}}\{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \in \mathbb{Q}_+\}, \\ \mathcal{N}_{\text{pr}} &= \overline{\text{span}}(\mathbb{R}\mathbf{1} + \mathcal{N}_{\text{int}}).\end{aligned}$$

The closures are in the weighted strict topology.

Lemma 1.1 (Platform-wise uniform Cesàro tightness). *On each platform, the chosen weight satisfies*

$$\sup_{T \geq 1} \frac{1}{T} \int_0^T \int W \circ \Theta_t d|\mu| dt \leq C \int (1 + W) d|\mu|$$

for every weighted finite signed measure μ . Consequently the Cesàro averages of $(\Theta_t)_\# \mu$ are uniformly weighted-tight.

Proof. For the compact Sinai platform, the path weight is controlled by the exponential return/roof Lyapunov function, whose expected drift is negative outside a compact set. For hard spheres, mass and energy are conserved and the contact-count weight has a uniform short-time exponential moment from B2. Integrating the respective Foster–Lyapunov inequalities yields the displayed uniform averaged bound. Weighted Prokhorov compactness gives tightness. $\square$

Theorem 1.2 (Invariant separator and Hausdorff cotangents). *A potential has zero integral under every invariant weighted probability if and only if it belongs to $\mathcal{N}_{\text{int}}$. Two potentials have the same long-time pressure if and only if their difference belongs to $\mathcal{N}_{\text{pr}}$. Both quotients are Hausdorff, and their continuous duals are countably additive invariant Radon measures with the appropriate normalization.*

Proof. Coboundaries integrate to zero and constants shift pressure. Conversely, separate a class outside the closed linear span by a weighted Radon functional using the strict dual theorem. Average that functional over rational times. Lemma 1.1 gives a weighted-tight subsequence; its limit is invariant and preserves the separating pairing because the averaged coboundary error is an endpoint term divided by time. Jordan decomposition and normalization give invariant probabilities. The pressure statement follows from the variational principle and the constant direction. $\square$

1.2. A common transfer Banach bundle. For a regular phase chart $\eta \mapsto \mathcal{L}_\eta$, all transfer operators act on the common strong/weak Banach bundle furnished by A1 or A2, not on mutually identified $L^2(\nu_\eta)$ spaces. Let Π_η be the simple Riesz projector and S_η its reduced resolvent.

Theorem 1.3 (Kato transport without infinite-path Radon–Nikodym densities). *The differential equation*

$$U'_\eta = [\Pi'_\eta, \Pi_\eta]U_\eta, \quad U_0 = I,$$

defines bounded Kato parallel transport on the common transfer bundle and satisfies $U_\eta \Pi_0 = \Pi_\eta U_\eta$. Covariant derivatives of the leading eigenfunction,

eigenfunctional, pressure, spectral projector and any finite resolved subspace are finite sums of $\mathcal{L}'_\eta$, S_η , and their iterates. No Radon–Nikodym derivative between distinct infinite-volume ergodic path measures is used.

Proof. The Riesz formula on the common Banach scale gives a continuously differentiable finite-rank projector. Kato’s commutator equation has a unique bounded evolution and differentiating $U_\eta^{-1}\Pi_\eta U_\eta$ shows it is constant. Differentiating the eigen-equations and applying the reduced resolvent gives the response words. All objects remain in the common transfer space even when the corresponding stationary path measures are mutually singular. $\square$

1.3. Resolved memory on the transported subspace. Let $P_\eta = U_\eta P_0 U_\eta^{-1}$ be the transported finite resolved projection in the common Hilbert realization of the A4 Doob phase. Put

$$C_\eta(z) = P_\eta(z - L_\eta)^{-1}P_\eta, \quad \widehat{K}_\eta(z) = zP_\eta - P_\eta L_\eta P_\eta - C_\eta(z)^{-1}.$$

Theorem 1.4 (Covariant compressed resolvent and memory). *The covariant derivative of C_η is the resolvent Duhamel insertion plus the two transported projection derivatives. Differentiating the displayed identity gives the covariant memory response, including derivatives of the phase generator and resolved subspace. It agrees with first differentiating the A4 history pressure and then compressing.*

Proof. Differentiate $P_\eta(z - L_\eta)^{-1}P_\eta$ in the common bundle and use $D(z - L)^{-1} = (z - L)^{-1}(DL)(z - L)^{-1}$. Matrix inversion gives $D(C^{-1}) = -C^{-1}(DC)C^{-1}$. The A4 pressure tangent is the same propagated correlation operator before Laplace transformation, so Fubini and normal resolvent bounds prove the commutation. The sign is the causal Volterra sign of A4. $\square$

1.4. Optional projections and nonlinear contractions.

Theorem 1.5 (Stable optional-projection limit). *Finite deterministic likelihood martingales projected onto the A4 past filtration converge stably, jointly with the enhanced path tangent, to the optional projection of the limiting diffusion likelihood. The result holds for every bounded terminal cylinder and every regular phase chart.*

Proof. A4 gives locally uniform convergence of conditional future kernels on compact past histories and uniform exponential moments. Apply the kernels to the finite terminal likelihoods and use stable convergence of the enhanced path. Uniform integrability permits conditional expectations to pass to the limit. The limit is the diffusion optional projection by uniqueness of regular conditional expectation. $\square$

Theorem 1.6 (Typed nonlinear contraction and tangent chain rule). *Let $\mathcal{C} : X \rightarrow Y$ be a continuous nonlinear observable map with Fréchet derivative on the regular rate sublevel. The contracted pressure is*

$$Q_{\mathcal{C}}(\varphi) = Q(\varphi \circ \mathcal{C}),$$

not $Q(\mathcal{C}^* \varphi)$ for a fictitious nonlinear adjoint. At an exposed phase x_* ,

$$DQ_{\mathcal{C}}(\varphi)h = DQ(\varphi \circ \mathcal{C})[D\varphi(\mathcal{C}x_*)D\mathcal{C}(x_*)h],$$

and the Hessian contains the covariance pullback plus the second derivatives of φ and $\mathcal{C}$. The rate is the usual nonlinear contraction $I_{\mathcal{C}}(y) = \inf_{\mathcal{C}x=y} I(x)$.

Proof. The finite identity $e^{\mu\varphi(\mathcal{C}(X_{\varepsilon}))}$ is exactly the original source functional composed with $\mathcal{C}$. The contraction principle gives the rate. Fréchet chain rules on the regular chart give the first and second responses, while the covariance term is supplied by A4/B3. This formulation is typed for a nonlinear map and requires no nonlinear linear adjoint. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`